

Rising Beyond Borders

Document 26 — Production Host & Infrastructure Configuration

Master Specification + Claude Code Implementation Prompt

Status: Planning/configuration phase. This document remains host-neutral until RBB selects a production host and domain. It does not authorize production launch.

1. Purpose

Connect the deterministic release process established in Document 25 to a real production hosting environment without changing application architecture unnecessarily. Host-specific configuration must remain the smallest possible adapter.

Do not invent missing RBB content, credentials, DNS records, infrastructure ownership, or hosting choices.

2. Current Starting State

- Application is STAGING-READY, not production-ready.
- Production builds are demo-free and protected by the release gate.
- Release packaging is deterministic and refuses staging builds.
- A repeatable smoke test can target a real URL.
- Forms, donations, analytics and monitoring remain disabled.
- No production domain, host, DNS or TLS configuration has been selected.

3. Decisions Required Before Host Configuration

- Production domain.
- Hosting platform/provider.
- DNS owner and access.
- Hosting account/billing owner.
- Deployment access and deployment method.
- Preview/staging deployment model.
- Operational/incident contact.
- Whether host-injected analytics, cookies, monitoring or support widgets are permitted.

4. Host Requirements

Requirement	Required behavior
HTTPS	Serve production over HTTPS; permanently redirect HTTP.
Redirects	Legacy redirects are real permanent HTTP redirects.
404	Unknown URLs return a true 404 response.
HTML	Pre-rendered route HTML remains available.
Headers	Document 20 security headers remain effective.
CSP	Existing strict CSP is preserved; Razorpay origins only when enabled.
Secrets	Server secrets never enter client bundles.
Logs	Operational logs available without unnecessary personal data.

5. Domain, DNS & TLS

- Use only the final RBB-approved domain.
- Record DNS provider and ownership in docs/DEPLOYMENT.md.
- Use only DNS records supplied by the selected host; do not invent IPs/CNAMEs.
- Document www/non-www canonical behavior.
- Verify the final domain serves the intended release, not a default host page.
- Verify certificate chain and HTTPS externally.
- Verify HTTP→HTTPS permanent redirect.
- Enable HSTS only after HTTPS is confirmed stable; do not preload without an explicit infrastructure decision.
- Verify no mixed-content requests.

6. Routing & Redirects

- Serve valid pre-rendered application routes.
- Unknown routes must produce a real 404.
- Move SEO-critical legacy redirects to host-level 301/308 rules.
- Preserve canonical trailing-slash/query/hash behavior.
- Do not depend solely on client-side redirects.

7. Static Assets & Caching

- Cache hashed immutable assets aggressively where safe.
- Never publicly cache form submissions or payment API responses.
- Ensure HTML caching does not serve stale environment-specific content after release.
- Document CDN/cache invalidation behavior.
- Verify all WebP/responsive image assets remain accessible.

8. Production Environment

- Set VITE_SITE_URL to the actual HTTPS production origin.
- Keep server secrets only in host-side secret storage.
- Keep EMAIL_MODE off until the Resend checklist is complete.
- Keep DONATION_ENABLED false until Razorpay approval/configuration is complete.
- Keep measurement streams disabled until separately approved.
- Never copy staging .env files into production.

9. Resend

- Use a newly rotated API key; never reuse the previously exposed key.
- Verify RBB sending domain, From address and approved recipients.
- Keep forms disabled until Privacy Policy and recipient approvals are complete.
- Verify server-side delivery from the real runtime before enabling live forms.
- Never expose the key in Vite/client code, logs or artifacts.

10. Razorpay

- Host the existing donation endpoints without rewriting their security behavior.
- Keep key secret and webhook secret server-side.
- Verify HTTPS webhook reachability and POST handling.
- Preserve signature verification and idempotency.
- Do not enable live donations without production credentials and RBB approval.

11. Rate Limiting & Runtime

Document 25 requires live forms/donations to fail closed unless the runtime provides an approved shared limiter or a confirmed single long-running process.

- Identify whether the host is serverless, edge, container-based or long-running.
- Document scaling behavior.
- For multi-instance/serverless deployment, configure the approved shared limiter before live activation.
- Use single-instance mode only after the host architecture is explicitly verified.
- Do not treat a local in-memory limiter as production-safe on a multi-instance host.

12. Real-Host Security Audit

- Compare response headers against docs/SECURITY.md.
- Verify CSP in enforce mode.
- Verify HSTS after HTTPS.
- Verify frame protections, X-Content-Type-Options, Referrer-Policy and Permissions-Policy.
- Verify the host does not inject scripts that violate CSP or privacy assumptions.

13. Third-Party Technology Audit

- Run the existing technology audit against the real host.
- Compare third-party requests against the approved inventory.
- Check host cookies and browser storage.
- Check host analytics, performance scripts, support widgets and injected resources.
- Update privacy/technology documentation only after real host behavior is known.

14. Preview Deployment

- Deploy the exact release package or deterministic equivalent.
- Use a non-indexable preview URL.
- Run `npm run smoke -- --url=`.
- Run browser smoke checks.
- Check headers, 404s, redirects, assets, CSP, cookies/storage and third-party requests.
- Resolve host-specific problems before production DNS cutover.

15. Production Cutover

- Confirm release candidate and build digest.
- Configure production environment variables/secrets.
- Deploy the verified release package.

- Verify deployment before DNS cutover.
- Point the production domain to the verified deployment.
- Verify HTTPS, redirects and headers.
- Run the full smoke test against the real production URL.
- Verify robots, sitemap and canonicals.
- Confirm forms/donations remain in their intended disabled state unless separately approved.
- Record deployment revision and time.

16. Rollback

- Keep the previous known-good release package.
- Rollback by redeploying the previous artifact; do not manually edit production files.
- Run smoke tests after rollback.
- Verify DNS/TLS/security behavior remains correct.
- Record rollback reason and resulting release revision.

17. Host Acceptance Checklist

- HTTPS PASS
- HTTP→HTTPS PASS
- Permanent redirects PASS
- True 404 PASS
- Security headers PASS
- CSP PASS
- Canonical URLs PASS
- robots.txt PASS
- sitemap.xml PASS
- Images/assets PASS
- Hydration/browser behavior PASS
- Cookies/storage PASS or explicitly approved
- Third-party requests expected only
- Demo-content scan PASS
- Release digest recorded PASS

18. Do Not Do

- Do not choose a host by assumption.
- Do not add host dependencies before the host is selected.
- Do not put secrets in frontend/Vite variables.
- Do not enable forms or donations merely because their code exists.
- Do not add analytics without approval.
- Do not publish staging/demo content.
- Do not weaken CSP to solve a host issue without identifying the exact request.
- Do not claim production readiness before real-host smoke testing passes.

19. Claude Code Implementation Prompt

You are implementing Document 26 — Production Host & Infrastructure Configuration for Rising Beyond Borders.

Treat Documents 1–25 and the current repository as authoritative.

The production host, domain, DNS and TLS provider have NOT been selected. Do not invent them.

Before host-specific changes:

1. Audit the deployment package, smoke test, release gate, security, forms/email, donations and deployment docs.
2. Report what cannot be configured until RBB selects the host.
3. Keep the application host-neutral.

Once a real host is supplied:

4. Add the smallest possible host adapter.
5. Preserve pre-rendered HTML, real 404 behavior and permanent redirects.
6. Preserve strict CSP/security headers.
7. Preserve production/staging separation.
8. Keep Resend/Razorpay secrets server-side.
9. Keep forms/donations gated.
10. Configure rate limiting according to the actual runtime model; fail closed when shared/single-instance requirements are unmet.
11. Configure HTTPS, routing, caching and redirects for the selected host.
12. Deploy a preview before production.
13. Run `npm run smoke -- --url=` plus browser checks.
14. Audit cookies, storage, third-party requests and host-injected scripts.
15. Run the final smoke test only after production DNS points to the verified deployment.
16. Do not stage or commit.

Final report:

- host/provider
- domain
- DNS/TLS ownership
- host configuration
- environment variable names only
- runtime/scaling model
- rate-limit configuration
- preview URL
- smoke/browser results
- headers/CSP result
- cookies/storage/third-party audit
- production URL if deployed
- rollback procedure
- unresolved RBB decisions/blockers

Do not claim production-ready until the real host deployment and smoke test pass.

20. RBB Infrastructure Approval Matrix

Decision	Owner	Status
Production domain	RBB	Pending
Hosting provider	RBB	Pending
DNS access/ownership	RBB	Pending
TLS/HTTPS	RBB + host	Pending
Deployment access	RBB + engineering	Pending
Incident contact	RBB	Pending
Resend production	RBB	Pending
Razorpay production	RBB	Pending
Analytics/monitoring	RBB	Pending

21. Definition of Done

- Real host and domain selected and documented.
- DNS/TLS ownership known.
- Preview deployment passes smoke/browser checks.
- Host headers and CSP verified.
- Permanent redirects and true 404 verified.
- Production artifact remains deterministic and demo-free.
- Rate limiting matches actual runtime architecture.
- Forms/donations remain safely gated.
- Production deployment passes complete smoke test.
- Rollback is executable.

End of Document 26